

WBS Final Project — Project Proposal

Explainable Machine Learning for Anomaly Detection and Fault Diagnosis in Batch Distillation

1. Project Overview

The project aims to develop an **explainable machine-learning approach for detecting anomalies and diagnosing process faults in a batch distillation process**.

The project will use a publicly available dataset containing **real experimental process data** from a laboratory-scale batch distillation plant, including sensor measurements, actuator signals, and detailed anomaly information.

The goal is not only to build a predictive model, but to investigate whether ML can provide **interpretable information about why a process is classified as abnormal**.

2. Problem Statement

Chemical and industrial processes can experience disturbances caused by factors such as:

- sensor problems,
- actuator disturbances,
- changes in process conditions,
- disturbances affecting process operation.

Detecting that a process is abnormal is useful, but process engineers also need to understand:

What is causing the abnormal behavior, and which process variables provide evidence for this diagnosis?

This project therefore combines **anomaly detection, fault diagnosis, and explainable ML**.

3. Dataset

The project will use a publicly available **Batch Distillation Dataset from RPTU/TUM**.

Key characteristics include:

- **119 experiments**
- Real experimental process data
- Measurements at approximately **1-second intervals**
- Process sensor measurements
- Actuator measurements
- Normal and anomalous experiments
- Detailed anomaly/fault metadata
- Additional NMR, GC and simulation data available for potential future extensions

The dataset provides a strong connection between **chemical engineering/process engineering and machine learning**.

4. ML Approach

The project will follow a **progressive three-level approach**.

Level 1 — Anomaly Detection

First, investigate:

Can ML distinguish normal from abnormal process behavior?

Process Data
↓
Feature Engineering
↓
Classification Model
↓
Normal / Anomaly

This forms the **core project objective**.

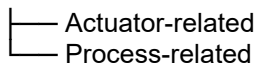
Level 2 — Fault Diagnosis

If Level 1 is successful, investigate:

Can ML distinguish different types of process disturbances?

For example:

Anomaly
↓
Fault Category
|—— Sensor-related



This is an important extension, but it will not compromise the core project if the available data does not support it sufficiently.

Level 3 — Explainable ML

The main differentiating component:

Which process variables drive the model's predictions?

Methods such as **SHAP** and **feature importance** will be used to investigate the model's decisions.

The resulting explanations will then be compared with the **known/documentd causes of the process disturbances**.

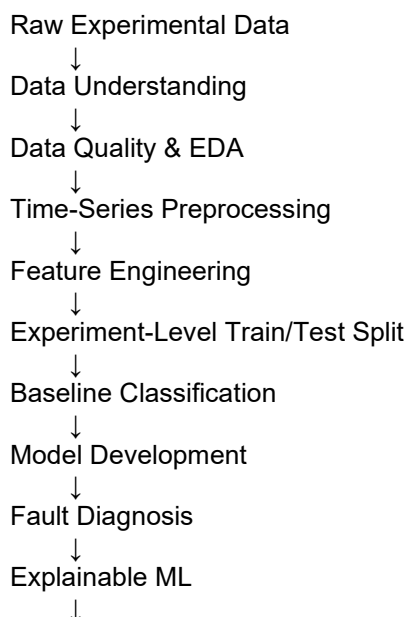
This allows us to ask not only:

Is the model accurate?

but also:

Does the model identify process variables that make sense from a chemical/process-engineering perspective?

5. Planned Workflow



Process Engineering Interpretation



Final Evaluation

A key methodological consideration will be avoiding **data leakage between experiments**. The train/test split will therefore be performed at the experiment level rather than randomly splitting individual time-series observations.

6. Planned Models

The exact model selection will be determined after EDA and feature engineering.

Initial candidates:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting / XGBoost

The models will be compared using appropriate classification metrics, rather than relying on accuracy alone.

Potential metrics include:

- Precision
 - Recall
 - F1-Score
 - Balanced Accuracy
 - ROC-AUC / PR-AUC
 - Confusion Matrix
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7. Explainability

The explainability stage will investigate:

- Which process variables are most important?
- Are important variables consistent across experiments?
- Which variables influence individual predictions?
- Do the model explanations correspond to documented fault mechanisms?
- Can ML results be interpreted in a meaningful process-engineering context?

This is particularly important because the project aims to demonstrate **interpretable industrial ML**, rather than treating the model as a black box.

8. Project Scope

Core — guaranteed

Anomaly Detection + Explainable ML

Target — if supported by the data

Fault Diagnosis

Out of scope for the initial implementation

- Deep-learning/time-series architectures
- NMR/GC integration
- Simulation integration
- Real-time deployment
- Web application
- Production deployment

These can be documented as **potential future work** rather than risking the core project timeline.

9. Timeline

The project has a fixed presentation deadline of **17 September 2026**.

We have approximately **four weeks** available.

Phase	Dates	Main outcome
1. Data & Problem Definition	20–26 Aug	Dataset understood and ML problem defined
2. EDA & Feature Engineering	27 Aug–2 Sep	ML-ready dataset + first model
3. ML Development	3–9 Sep	Final candidate model
4. Explainability & Validation	10–14 Sep	Final analysis
5. Finalization	15–16 Sep	Documentation + presentation
Final Presentation	17 Sep	Project presentation

Key milestones

26 Aug — M1: Data Ready

Dataset, target and evaluation strategy defined.

2 Sep — M2: First Model

Complete first pipeline from raw data to prediction.

9 Sep — M3: Model Freeze

Final ML approach selected.

14 Sep — M4: Analysis Freeze

Explainability and technical analysis completed.

16 Sep — M5: Project Freeze

Final project and presentation ready.

17 Sep — Final Presentation

10. Expected Deliverables

By the end of the project:

1. **Clean and reproducible data-processing pipeline**
 2. **Exploratory data analysis**
 3. **Feature-engineering pipeline**
 4. **Baseline and final ML models**
 5. **Model evaluation and comparison**
 6. **Fault-diagnosis analysis**, if supported
 7. **SHAP / Explainable ML analysis**
 8. **Process-engineering interpretation**
 9. **Final documentation / GitHub project**
 10. **Final WBS presentation**
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11. Why This Project

The project combines three areas directly relevant to my background and target career:

Chemistry

→ understanding chemical processes and separation

Process Engineering

→ sensors, actuators, process behavior and fault mechanisms

Data Science

→ Python, feature engineering, machine learning, model evaluation and Explainable AI

This creates a direct bridge between my existing scientific/engineering background and my transition toward **Data Science, Analytics Engineering and Industrial Data & AI**.

12. Final Project Objective

Develop and evaluate an explainable machine-learning approach that can detect abnormal behavior in a batch distillation process and, where supported by the data, identify the underlying fault type and the process variables driving the model's decision.

The project will therefore demonstrate not only **predictive performance**, but also whether machine-learning results can be translated into **meaningful process-engineering insights**.